Expansion and Standardization of Rat Expression Data at the Rat Genome Database

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https://rgd.mcw.edu

Rat Genome Database

Abstract

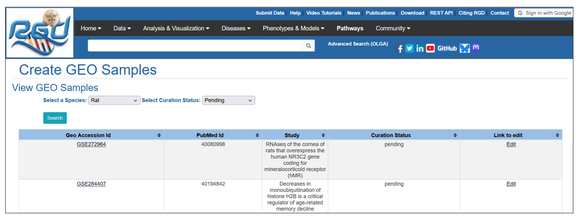
Data sharing is a fundamental pillar of scientific research that promotes data reproducibility, transparency, and preservation. The practice of mining of public data repositories designed to support this pillar serves as a core component in accelerating scientific research due to the quantity and comprehensiveness of the available data. The Rat Genome Database (RGD) began incorporating RNA-seq expression data for supported species (human, rat, pig, vervet, dog) from the Expression Atlas (EBI) and the Gene Expression Omnibus (GEO) into gene report pages in 2019. Rat data imports were limited to studies based on untreated controls and data aligned to Rnor6.0 assembly with data type TPM/FPKM, resulting in the import of seven studies. RGD is expanding and standardizing RNA-seq expression data, and integrating into the broader RGD ecosystem, guided by three primary objectives: (1) metadata standardization with Natural Language Processing (NLP)-assisted curation, (2) quantitative expression data standardization, and (3) increased accessibility of expression data through the RGD user interface.

Public data repositories such as GEO adhere to Minimum Information About a Sequence Experiment (MINSEQE) standards, which capture experimental descriptors in a flexible, non-standardized manner that supports experiment mining but limits specific criteria-based identification. As a result, researchers may have difficulty finding studies of interest in the context of tissue, rat strain, or experimental condition. To address this challenge, RGD developed an NLP-assisted manual curation tool to aid curators in the standardization of experimental metadata for expression studies deposited in public repositories. The tool provides six controlled vocabulary fields: UBERON: Cross-Species Anatomy, VT: Vertebrate Trait, CMO: Clinical Measurement, CL: Cell, RS: Rat Strains, and XCO: Experimental Conditions. Terms are prepopulated where the NLP prediction algorithm identifies a match for that category. Terms are verified or edited by curators, while missing information is manually gathered from the GEO Accession Display and available associated publications.

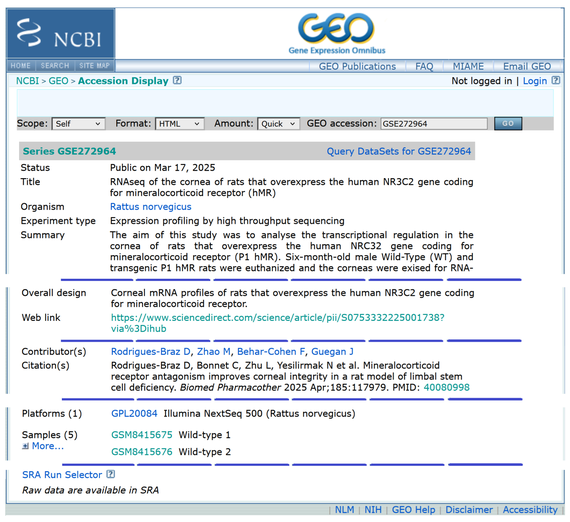
The development of the NLP-assisted curation tool facilitates efficient data mining and has expanded RGD's ability to import both control-only and experimental *in vivo* studies. However, data in public repositories represent experiments conducted over time and thereby are generated with many combinations of sequencing technologies, genome assembly versions, and bioinformatic workflows adding complexity to the ability to compare quantitative data across studies. In response to this challenge, RGD designed an RNAseq workflow utilizing the STAR and RSEM algorithms to produce standardized expression values aligned to the mRatBN7.2 assembly. These data are subsequently displayed on the gene report pages and allow for comparisons across studies.

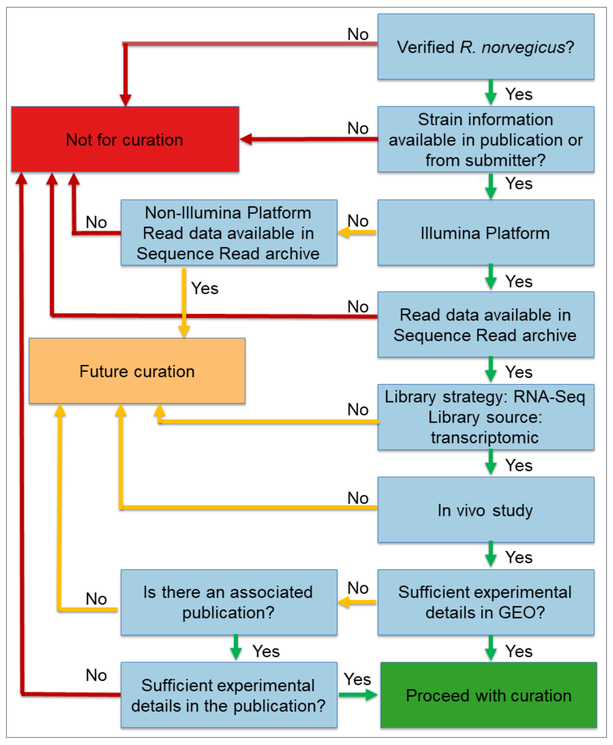
The development of the NLP-assisted curation tool and RNAseq workflow has significantly increased expression data content within the RGD ecosystem. Metadata and quantitative data for over 500 studies are searchable within the database, with additional gene and transcript data downloads and interactive expression profile displays planned in an upcoming release.

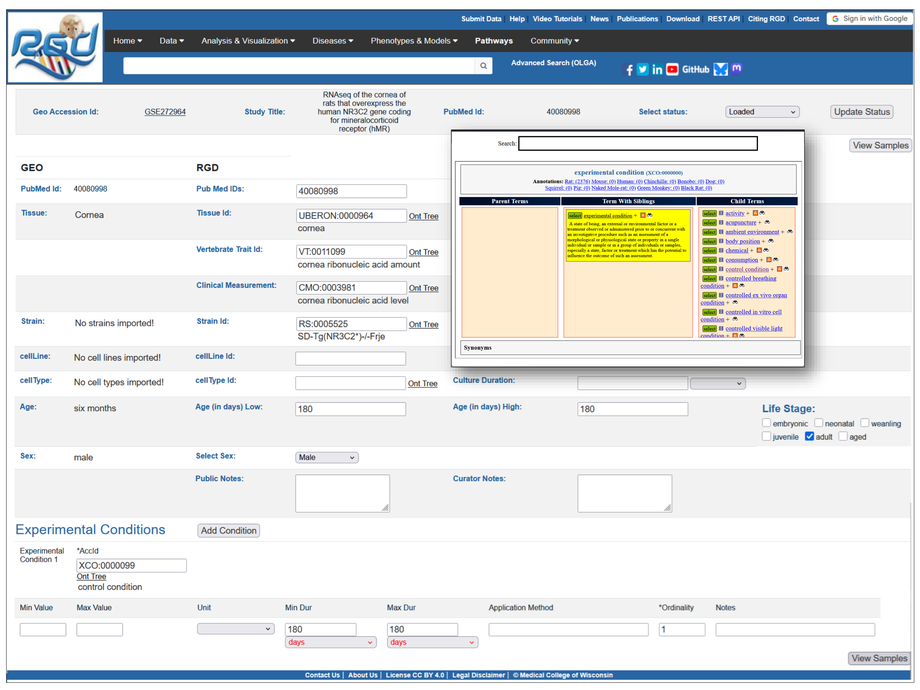
Objective 1: Metadata Standardization



Transcriptomic studies are loaded into the RGD expression tool (above). Curators review the information presented in the GEO Accession Viewer (below) and determine if a study meets the necessary criteria to be considered for curation (right).

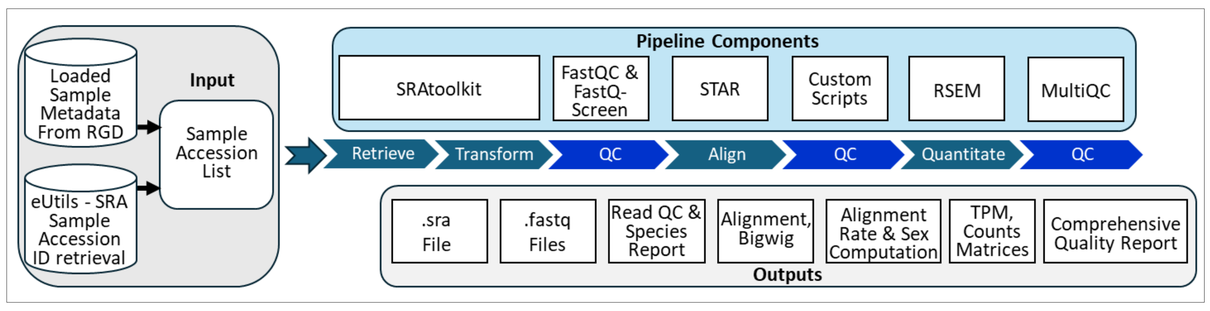






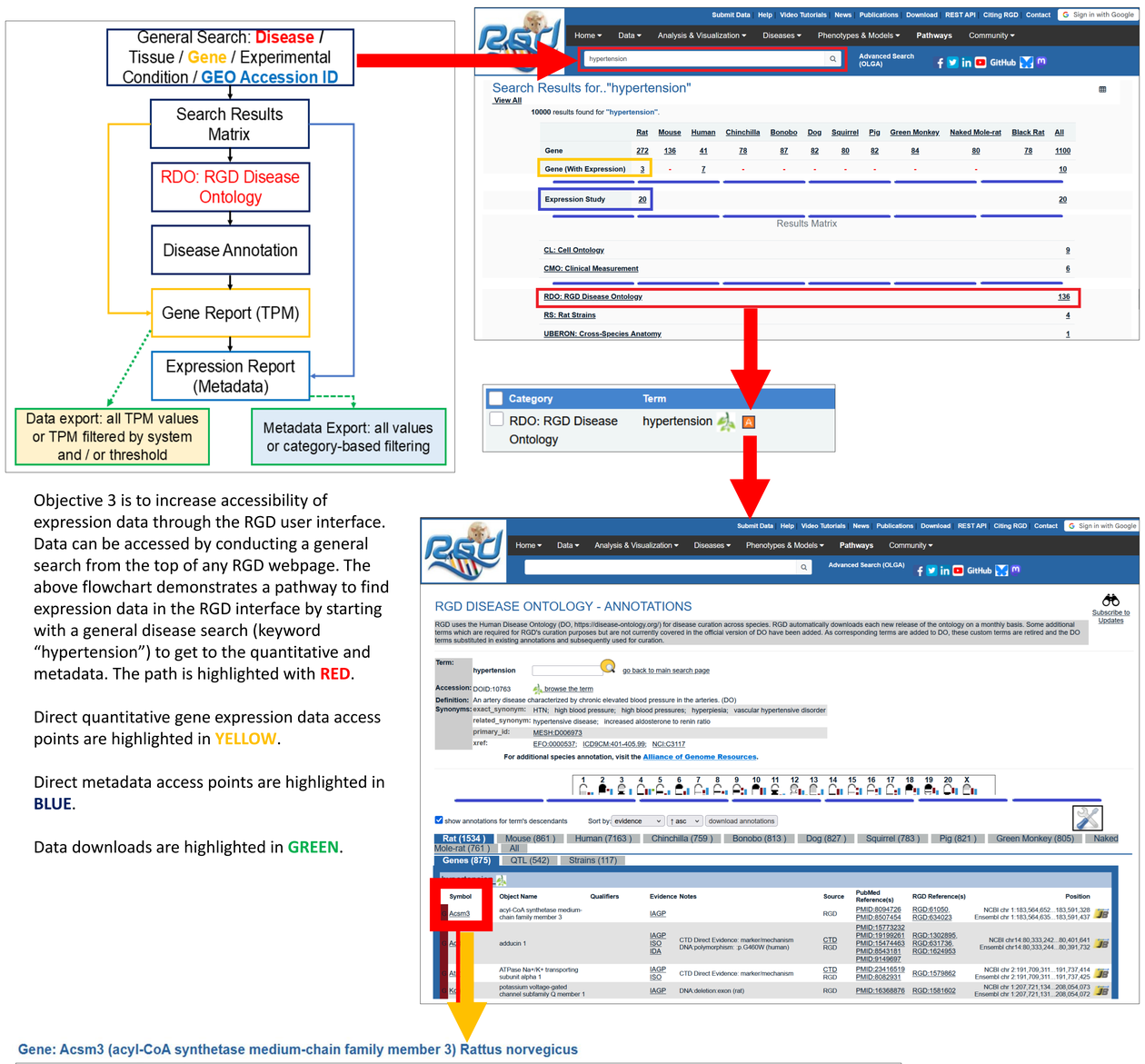
Metadata are extracted from the imported GEO study SOFT files. The NLP algorithm auto-populates UBERON, CMO, VT, Strain, and Cell Ontology terms. Curators add, accept, or override terms as needed. Pop-up ontology browsers (inset) are available for each field. Studies are first curated at the study level and terms are propagated to all applicable samples. Curators then modify individual sample attributes based within the next window of the interface prior to loading metadata to the underlying data tables. In the above example the PMID of the provided article, UBERON, VT, and CMO and sex were auto-populated based on the NLP algorithm. All other data were entered by the curator.

Objective 2: Quantitative Expression Data Standardization



The RGD RNAseq reprocessing pipeline utilizes publicly available software packages to retrieve & transform raw data from the SRA repository. STAR Aligner and RSEM are used for alignment and expression abundance estimation. Currently, the pipeline is designed to process raw reads generated on the illumina sequencing platform.

Objective 3: Data Access in RGD User Interface

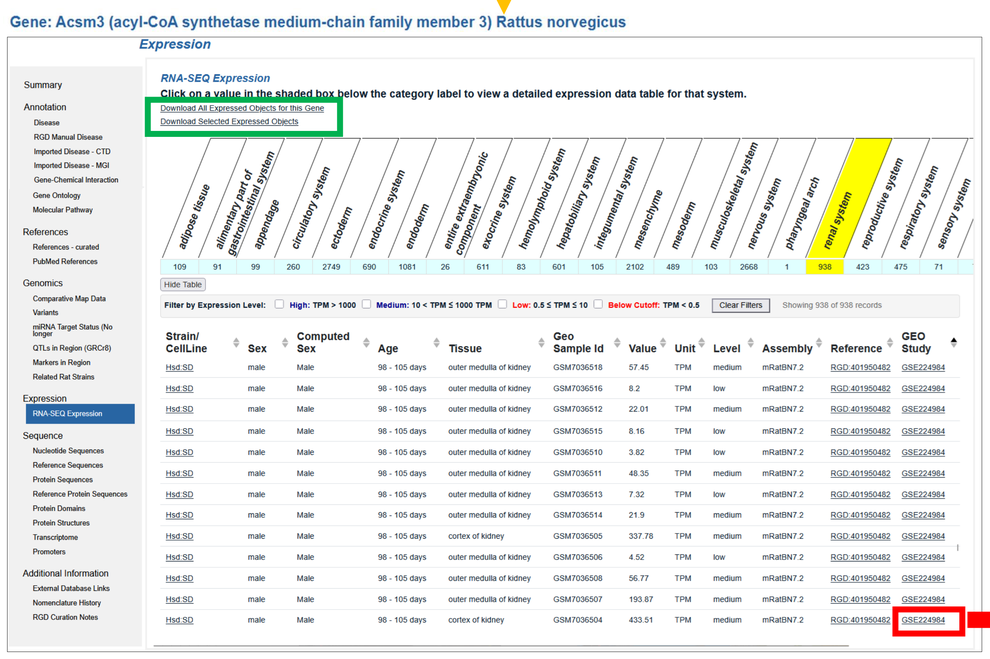


Objective 3 is to increase accessibility of expression data through the RGD user interface. Data can be accessed by conducting a general search from the top of any RGD webpage. The above flowchart demonstrates a pathway to find expression data in the RGD interface by starting with a general disease search (keyword "hypertension") to get to the quantitative and metadata. The path is highlighted with **RED**.

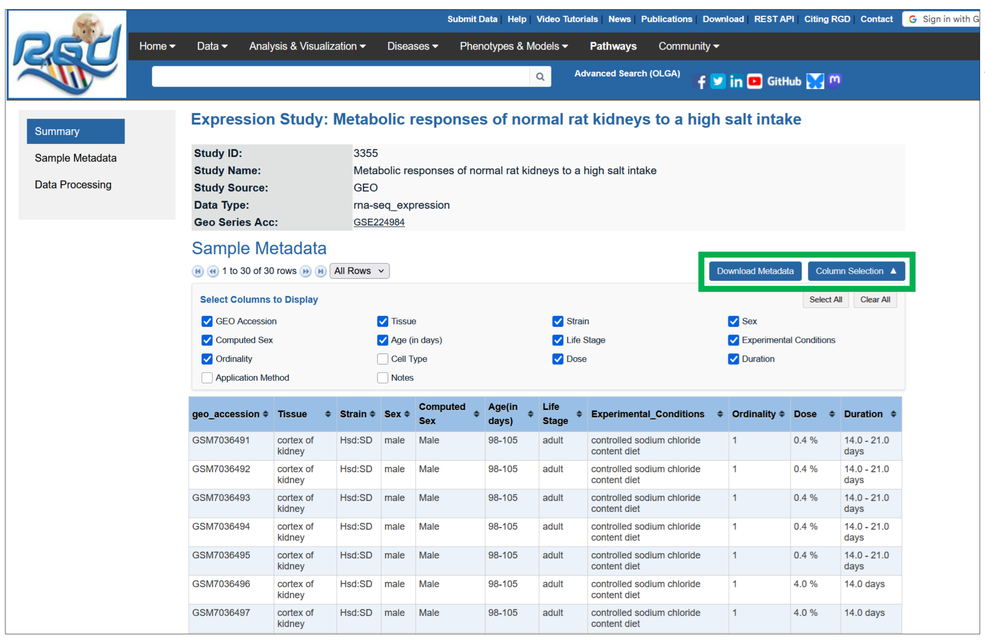
Direct quantitative gene expression data access points are highlighted in **YELLOW**.

Direct metadata access points are highlighted in **BLUE**.

Data downloads are highlighted in **GREEN**.

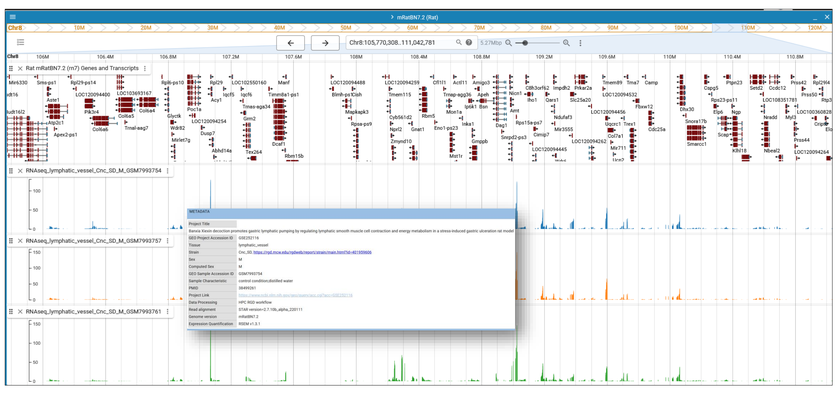


Alternatively, this Gene Report Page (left) can be reached by starting with a search for a gene of interest. The Expression section is easily navigated to via the left-hand menu bar. Users can filter data by system, tissue, or expression threshold. All or filtered data can be downloaded for further analysis.



Curated Metadata for a study is accessible via a search for a specific study ID, an experimental condition, tissue, or disease. Users then select the "Expression Study" results from the search matrix and navigate through the results list to get to the metadata of interest. Users can use the checkboxes in the Metadata Report Page (left) to view specific metadata categories. All or filtered data can be downloaded for further analysis.

Future Directions



- Upcoming GRCr8 TPM data to be added to Gene Report pages.
- Additional Gene and Metadata Report Page data visualizations.
- RNAseq workflow output downloads:
  - JBrowse2 coverage tracks (left)
  - Study-level data quality reports
  - Gene- and transcript- level TPM matrices
  - Gene- and transcript- level raw counts matrices
- Expression curation for other RGD supported species.
- Single-cell study curation.

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Medical College of Wisconsin · Kidneydata · GBC Global Core Biodata Resource · Alliance of Genome Resources Founding Member · National Heart Lung and Blood Institute

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